



Triple Tres (333)

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AI / LLM

MCP ProtocolAgentic Loops

LLM Orchestration

MCP PrimitivesMulti-Agent

Claude APITool Use API

RAG · pgvectorSSE Transport

n8n FlowsLangfuseMastra

HITL PatternsMake · Zapier

BACKEND / INFRA

TypeScriptNode.jsPython

PostgreSQLSQLiteDocker

RailwayREST APIsGitLinux

FRONTEND

Next.jsReactTailwind

TypeScriptshadcn/ui

EDUCATION & CERTS

- Model Context Protocol: Advanced Topics
Anthropic · Oct 2025
- Claude Code in Action
Anthropic · Sep 2025
- Introduction to Model Context Protocol
Anthropic · Sep 2025
- AI Fluency: Framework & Foundations
Anthropic · 2025
- n8n Self-Hosted for Enterprises
n8n · Nov 2025
- Building Scalable Agentic Systems
DataCamp · 2025
- B.A. Graphic Design
UAA · 2013

LANGUAGES

SpanishNative

EnglishProfessional — technical documentation, proposals, and async collaboration with US-based teams

AI engineer and solo founder focused on replacing manual fiscal/accounting work with **production-grade agent systems**. Shipped a **43-tool MCP server**, multi-tier Claude orchestration, specialized agents and a full fiscal compliance SaaS; all from zero in TypeScript with strict Zod schema validations. Communicates and documents in **English-first environments** as a default.

EXPERIENCE

Founder & Lead AI Engineer · DISAI

Jun 2025 – Present

- 43-Tool MCP Server: Architected and deployed a MCP server (TypeScript/Node.js, Railway) covering all three primitives — **Tools** for CFDI 4.0 fiscal ops, **Resources** for live SAT catalog injection (batch-loaded per agent domain), SSE + HTTP Streamable dual transport, multi-tenant CSD management. EFOS/EDOS blacklist sync, PAC stamping. 100k+ invoices/min at <2ms latency. MCPTypeScriptRailway
- 3-Tier Agentic Orchestration: Haiku router (classifies 10 fiscal domains + query intent in ~100ms) → 10 domain DomainAgents (Sonnet, Anthropic native tool_use, real agentic loops up to 6 iterations, parallel tool calls, self-correction on failure) → Expert Registry with scoped tool allowlists (5–8/domain) + live MCP Resource injection. Zero hallucinations. Claude APITool UseMulti-Agent
- Vector DB & MCP Resource Injection: pgvector (PostgreSQL 16) for semantic SAT catalog search. MCP Resources pre-loaded into each agent’s context window before the first LLM call, eliminating hallucinated catalog codes without RAG query overhead. pgvectorPostgreSQL
- Self-Hosted AI Infra: Deployed self-hosted n8n with direct MCP client integration for SAT tool execution. Replaced Zapier/Make — 80% cost reduction, 400+ integrations, full data ownership. Deployed MVPs to Railway before scaling. n8nDocker
- Streaming React Frontend: Next.js 16 + shadcn/ui chat interface with SSE event pipeline — Activity Log, Thinking Steps, Skeleton bubbles for 30–60s async LLM waits. Compliance dashboard, EFOS monitor, CSD status, driver.js onboarding, per-call savings tracking. Next.jsReact 19Tailwind
- Langfuse Observability: Instrumented the full orchestrator with Langfuse tracing — every LLM call, tool invocation, model selection, token cost, and streaming event tracked end-to-end. Supports cost attribution per session and failure triage without log digging. LangfuseTracing
- Human-in-the-Loop Reviewer Surfaces: Built compliance dashboard and EFOS monitor as Next.js reviewer interfaces with one-click approval flows, MCP-apps for interactive responses in chat and per-call savings tracking — wired directly into the agent pipeline as the human approval layer. Next.jsHITLMCP-UI

Core Contributor / Lead Developer · RetailDAO

Jun 2024 – Jan 2026

- Financial Analytics Terminal: Full-stack “Bloomberg-style” dashboard aggregating real-time data from Federal Reserve, on-chain APIs and crypto exchanges via WebSockets. Rate-limiting, memory caching, and graceful degradation for a self-healing low-latency system. ReactWebSocketsAPIs
- Smart Contract Audits & Documentation: Led internal audits and built a token distribution visualization platform. Mitigated investor risk and drove a 30% increase in token valuation to all-time highs.

Crypto Research Analyst · Coinsider

Mar 2024 – Sep 2025

- Data-Driven Intelligence: Delivered 25+ peer-reviewed analytics reports (1,200+ pages) — SWOT analysis, investor risk models, and macroeconomic forecasting for Web3 ecosystems. Translated dense technical data into content generating +70k views.

KEY PROJECTS

SAT-MCP Server

43-tool MCP server for Mexican fiscal compliance. Full primitive set: tools, resources, prompts, elicitations, tasks + 8 MCP-UI mini-apps for in-chat fiscal rendering. SSE + HTTP Streamable. CFDI 4.0 lifecycle, EFOS/EDOS sync, PAC stamping, multi-tenant CSD. 3,960 tests · 91%+ coverage.

- TypeScript
- MCP SDK
- Node.js
- Railway
- SQLite

DISAI_Conta - 3-Tier Agent System

Haiku router + 10 domain DomainAgents (Sonnet, native tool_use, 6-iter agentic loops) + Expert Registry with scoped allowlists and live MCP Resource injection. Streaming SSE UI for Mexican SMEs.

- Next.js 16
- Claude API
- Tool Use
- MCP Client
- shadcn/ui

n8n AI Automation Stack

Self-hosted n8n with direct MCP client integration for SAT tool execution. Specialized sub-agents per CFDI type, 400+ integrations, context bundle architecture.

- n8n
- MCP
- LLM Agents
- Docker

Homelab AI Infrastructure

PostgreSQL 16 + pgvector for semantic SAT catalog search. Self-hosted Gitea, static IPs, Samba share. Powers the MCP Resource injection pipeline in production.

- PostgreSQL
- pgvector
- Linux
- Gitea
- Docker

Mastra + Langfuse Stack Evaluation

Evaluated Mastra framework for HITL patterns, agent workflows, and MCP connections. Identified tradeoffs vs. custom harness (Soniox TTS incompatibility, Langfuse A/B testing depth). Production stack uses custom orchestrator with pure Langfuse cloud for full observability control.

- Mastra
- Langfuse
- HITL
- Observability